

Unit-3

Memory Organization

A computer system uses different types of memory to store data and instructions. Since no single memory device can provide **high speed, large capacity, and low cost** simultaneously, computers use a **memory hierarchy**. The memory hierarchy organizes different memories according to **speed, capacity, cost, and access time**. Frequently used data is stored in faster memories, while less frequently used data is stored in slower but larger memories.

Memory Hierarchy

Memory hierarchy is the arrangement of different types of memory in a computer system in such a way that the CPU gets frequently used data from the fastest memory while rarely used data is stored in slower memory.

The hierarchy is based on the following characteristics:

- Speed
- Cost per bit
- Storage capacity
- Access time
- Frequency of use

Modern processors operate much faster than memory devices. If only one type of memory were used, the CPU would spend much of its time waiting for data. Memory hierarchy solves this problem by placing frequently used data in fast memory and less frequently used data in slower memory.

Advantages

- Increases overall computer performance.
- Reduces CPU waiting time.
- Provides faster data access.
- Balances speed and storage capacity.
- Reduces overall system cost.

Levels of Memory Hierarchy

1. Registers

Registers are very small storage locations inside the CPU. They temporarily hold instructions, addresses, and data currently being processed.

Characteristics

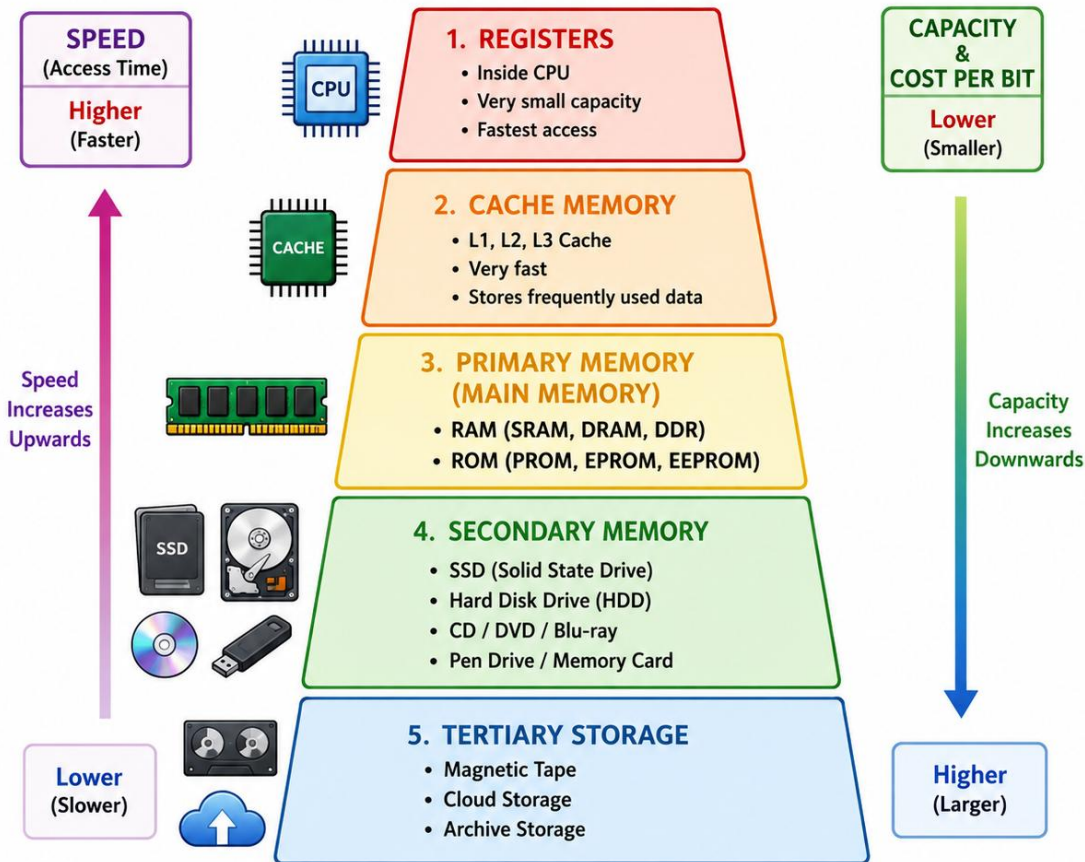
- Fastest memory
- Located inside CPU
- Very small capacity (Bytes)
- Highest cost per bit

Examples

- Program Counter (PC)
- Instruction Register (IR)
- Accumulator
- MAR
- MDR

MEMORY HIERARCHY

Arranged in order of speed, capacity and cost.



Key Characteristics



Speed :
Top is Fastest



Capacity :
Bottom is Largest



Cost per bit :
Top is Highest



Volatility :
Top (Registers, Cache, RAM)
are **Volatile**
(except ROM, SSD, HDD,
Tape, Cloud)

2. Cache Memory

Cache memory is located between the CPU and the main memory. It stores frequently used instructions and data.

Characteristics

- Faster than RAM
- Smaller than RAM
- Built using SRAM
- Reduces memory access time

Levels

- L1 Cache
- L2 Cache
- L3 Cache

3. Primary Memory

Primary memory stores programs and data currently in use.

It consists of

- RAM
- ROM

Characteristics

- Directly accessed by CPU
- Faster than secondary storage
- Moderate capacity
- Semiconductor memory

4. Secondary Memory

Secondary memory stores data permanently.

Examples

- Hard Disk
- SSD
- DVD
- CD
- USB Flash Drive

Characteristics

- Large capacity
- Non-volatile
- Low cost
- Slower than RAM

5. Tertiary Memory

Used mainly for long-term backup.

Examples

- Magnetic Tape
- Cloud Backup
- Optical Libraries

Characteristics

- Very large capacity
- Slowest access
- Cheapest storage

Principle of Locality

Memory hierarchy works because of the **Principle of Locality**, which states that programs tend to reuse the same data and instructions.

There are two types:

1. Temporal Locality

If a memory location is accessed once, it is likely to be accessed again soon.

Example: Variables inside a loop.

2. Spatial Locality

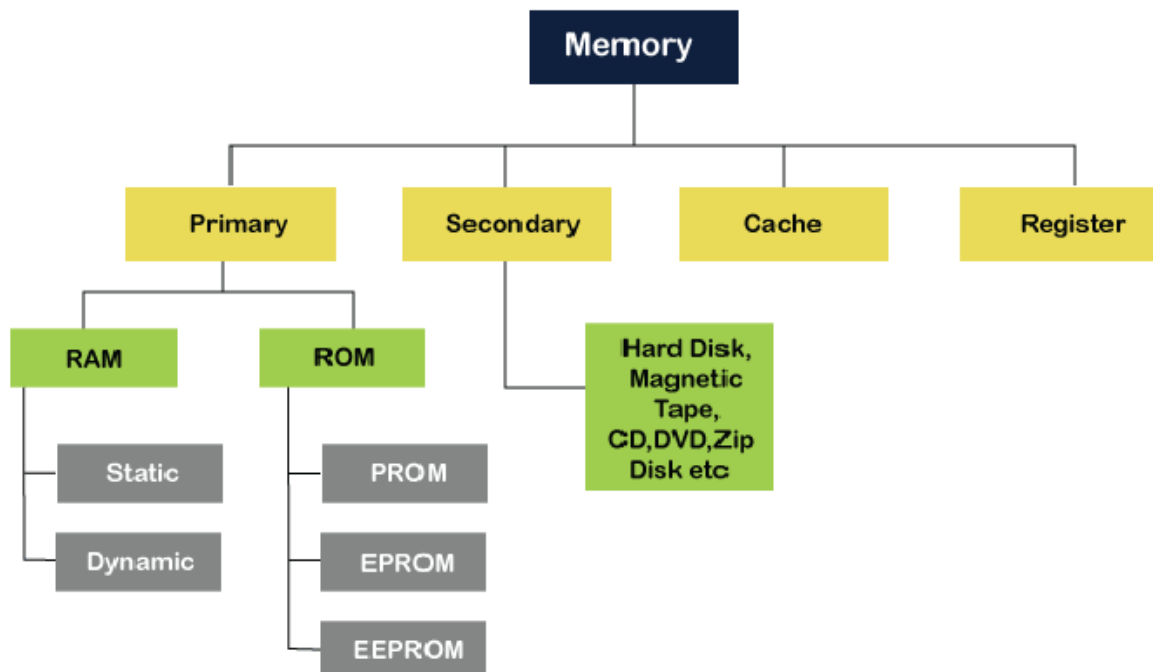
If one memory location is accessed, nearby locations are also likely to be accessed.

Example: Reading elements of an array.

Classification of Memory

Computer memory is classified into two major categories.

The following figure represents the classification of memory:



Types of Computer Memory

1. Primary Memory

Primary Memory, also known as **Main Memory** or **Internal Memory**, is the memory directly accessible by the **Central Processing Unit (CPU)**. It stores the data, instructions, and programs that the CPU is currently processing. Since the CPU can access primary memory quickly, it plays a vital role in determining the overall performance of a computer system.

Unlike secondary storage devices (such as hard disks and SSDs), primary memory provides **fast access** to information, allowing the processor to execute instructions efficiently.

Primary memory acts as the **working area of the computer**. Whenever a program is executed:

1. The operating system loads the program from secondary storage into primary memory.
2. The CPU fetches instructions from primary memory.
3. Data is processed.

4. Results are stored temporarily in primary memory before being saved to secondary storage.

Features

- Fast
- Semiconductor memory
- Directly accessible
- Limited storage
- Costly

Types

- RAM
- ROM

2. Secondary Memory

Secondary memory is a non-volatile storage device that stores data, programs, and information permanently. It provides large storage capacity at a lower cost than primary memory and retains data even when power is turned off.

Secondary memory has a **much larger storage capacity** than primary memory and is used to store the operating system, application software, documents, images, videos, music, databases, and backup files.

The CPU **cannot access secondary memory directly**. Whenever a program or data is required for processing, it is first transferred from secondary memory to **RAM (Primary Memory)** and then processed by the CPU.

Types

- Hard Disk
- SSD
- DVD
- Blu-ray Disc
- USB Drive
- Memory Card

Advantages of Secondary Memory

- Permanent data storage
- Very large storage capacity
- Cost-effective
- Suitable for backup and recovery
- Stores multimedia and large applications
- Portable storage devices are available

Primary Storage

RAM

Random Access Memory (RAM) is an essential type of computer memory that facilitates high-speed data storage and retrieval during active operations. Unlike sequential access memory, which processes data in a fixed order, RAM enables the CPU to directly access and manipulate

data stored at any physical location in memory. This capability eliminates unnecessary delays, making RAM used for real-time data processing and efficient system operations.

RAM is volatile memory, meaning that all stored data is erased when the power is turned off. This characteristic, while limiting its use for long-term storage, makes it ideal for temporary tasks that require immediate, rapid processing.

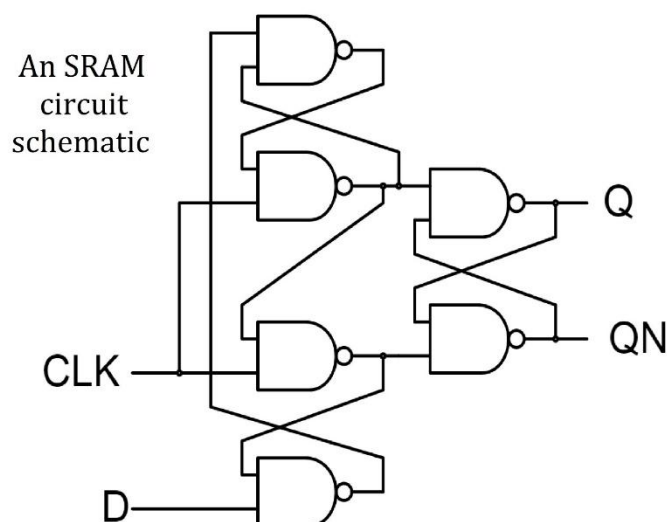
During operation, programs and active datasets are loaded into RAM, allowing the CPU to quickly fetch, modify, and store information without waiting for slower storage mediums like hard drives or SSDs. In practical terms, RAM acts as a buffer between the CPU and permanent storage. When you open an application, such as a web browser, the program's files and associated data are temporarily loaded into RAM. This ensures that actions like switching tabs, scrolling, or loading new content occur almost instantaneously.

Both SRAM and DRAM memory are volatile. This means that if they are disconnected from a power source, whatever information that they had in storage will eventually be lost. For SRAM this memory loss happens more or less instantaneously and for DRAM the memory will be lost within a few seconds.

RAM Types

1. SRAM (Static RAM)

SRAM uses flip-flop circuits to store each bit of data, providing high speed and reliability. It does not require periodic refreshing like DRAM, making it faster in accessing and retaining data as long as power is supplied. Due to its low latency and consistent performance, SRAM is used in applications where speed is critical, such as cache memory close to the CPU. Cache memory stores frequently accessed instructions or data, ensuring minimal delays during operations. For example, when you repeatedly execute a specific program, the CPU retrieves instructions almost instantaneously from SRAM, improving overall system efficiency. However, the complex design of flip-flop circuits increases its cost and reduces its storage density, limiting its use to smaller but high-performance tasks.



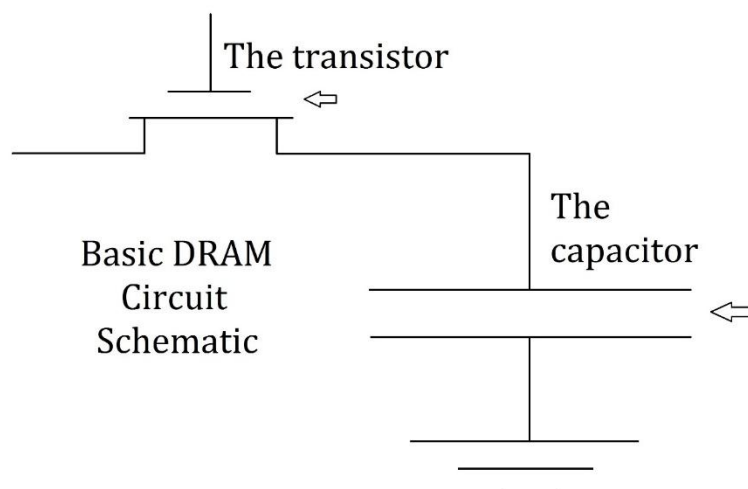
Basic Semantic of SRAM

Types of SRAM

- **Asynchronous SRAM:** Operates without a clock signal, relying on enable signals for read and write operations. It is commonly used in embedded systems, industrial devices, and microcontrollers due to its simplicity and suitability for lower-speed applications.
- **Synchronous SRAM:** Works in synchronization with a clock signal for faster and predictable memory access, making it ideal for CPU caches (L1, L2, L3) and high-performance processors in clock-driven systems.
- **Pipeline Burst SRAM:** Designed for high-speed operations with burst data transfers and overlapping tasks, it is widely used in networking equipment, graphics cards, and gaming systems for fast sequential data access.
- **Zero Bus Turnaround (ZBT) SRAM:** Eliminates delays between read and write operations by overlapping tasks, maximizing throughput in latency-sensitive environments like high-speed routers and real-time systems.
- **Non-Volatile SRAM (nvSRAM):** Combines the speed of SRAM with non-volatile storage to retain data during power loss, making it suitable for industrial automation, medical devices, and data logging applications requiring data persistence and quick access.

2. DRAM (Dynamic RAM)

DRAM stores data in capacitors, which require periodic refreshing due to charge leakage. This design is simpler and allows for higher storage density, making DRAM more cost-effective than SRAM. DRAM is commonly used as main memory, supporting larger datasets and running multiple applications simultaneously. For example, when you open multiple browsers tabs or edits large files, DRAM holds the data temporarily, enabling the CPU to access it as needed. Despite its slower speed compared to SRAM, DRAM's ability to store significant amounts of data at a lower cost makes it the backbone of general-purpose computing.



dynamic RAM

Basic schematic of

Types of DRAMS

- **Asynchronous DRAM (ADRAM):** Operates independently of the CPU's clock, relying on a dedicated memory controller to generate timing signals. Commonly used in older systems and simpler computing environments, it offers basic functionality but suffers from higher latency due to lack of synchronization.
- **Synchronous DRAM (SDRAM):** Synchronizes with the CPU's clock, ensuring precise timing for memory operations. Widely used in personal and enterprise computing, SDRAM eliminates wait states on the CPU-memory bus, improving efficiency during sequential data access.
- **Double Data Rate SDRAM (DDR SDRAM):** Enhances SDRAM performance by transferring data on both the rising and falling edges of the clock signal, effectively doubling the data rate. DDR SDRAM has evolved into multiple generations—DDR2, DDR3, DDR4, and DDR5—offering increasing speed, capacity, and efficiency for modern systems like gaming PCs, servers, and workstations.
- **Rambus DRAM (RDRAM):** Features a narrow, high-speed CPU-memory bus with advanced on-chip caching and precise signal timing. Known for its rapid data throughput, RDRAM is suited for applications like high-definition video processing and scientific simulations, though its high cost and proprietary nature limit widespread use.
- **Cache DRAM (CDRAM):** Combines on-chip SRAM as a high-speed cache for the slower DRAM cells. Frequently accessed data is stored in the SRAM cache, reducing latency and improving performance in repetitive tasks such as database queries and real-time analytics.

Comparison Between SRAM and DRAM

SRAM (Static RAM)	DRAM (Dynamic RAM)
Retains data as long as power is supplied using a flip-flop-based architecture.	Stores data in capacitors and requires periodic refreshing to maintain information.
Uses flip-flop circuits for each bit, ensuring stability without refresh cycles.	Stores data as electric charges in capacitors that require frequent refreshing to avoid loss.
Faster due to low access times and no refresh requirements, ideal for CPU caches.	Slower due to higher access times and refresh cycles, suitable for general-purpose memory.
Data remains stable as long as power is supplied.	Data life is short; capacitors leak charge and require constant refreshing.
Limited to 1 MB–16 MB due to complex circuit design.	Offers larger capacities, from gigabytes to terabytes, thanks to simpler design.

More expensive due to intricate flip-flop design, used in specialized applications.	Cheaper to produce, making it suitable for large-scale, cost-efficient memory solutions.
Consumes more power as flip-flops need a continuous power supply.	More energy-efficient but still consumes power due to periodic refresh cycles.
Performs reliably in extreme temperature environments, ideal for industrial or aerospace use.	Less resilient to temperature changes; performance may degrade in extreme conditions.
Complex design limits storage density and scalability.	Simpler design allows for higher storage density and better scalability.
Larger physical size due to lower packaging density.	Higher packaging density enables compact module designs for modern devices.
Commonly used in CPU caches (L1, L2, L3) for fast, low-latency operations.	Primarily serves as main memory for multitasking and running large applications.
Does not require refreshing, enabling uninterrupted and consistent performance.	Requires periodic refreshing, which introduces slight delays and complicates controller design.
Found in high-performance systems like CPU and GPU caches for critical speed-dependent tasks.	Widely used in main memory for devices like computers, smartphones, and servers.

DDR

As the speed of computer components have increased, memory speeds also improved. Double data rate (DDR) memory was introduced to consumers in 2000, prompting the previous technology to become known as single data rate (SDR).

DDR memory transfers data to the processor on both the rising and falling edges of the clock signal. This makes double data rate memory significantly faster – and more energy-efficient – than single data-rate memory, which uses only one edge of the clock signal to transfer data. Since DDR is faster and uses less energy than SDR, it quickly became the new standard for memory innovation.

DDR Generations

Each successive generation of DDR memory is faster and uses less energy than the one before it.

Today, most computers can get by using DDR3, while most high-end gaming and data-intensive creative content computers now use DDR4.

The latest generation, DDR5, was released in 2021 and offers next-level performance for serious gamers and professional creatives.

It important to determine what generation of memory your computer supports before buying new RAM. DDR5 memory, for example, will not fit on a motherboard built to support DDR4. Check with your motherboard manufacturer to ensure you have the right memory. Or, easier yet, use the Crucial System Scanner or System Selector to find guaranteed compatible memory for your computer.

Generation	Data rate (MT/s) Million transfers per second	Clock speed (MHz)	Voltage	Capacity per module	Pins
DDR1	200–400	100–200	2.5V	Up to 1GB	184
DDR2	400–800	200–400	1.8V	Up to 4GB	240
DDR3	800–2133	400–1066	1.5V	Up to 16GB	240
DDR3L	800–2133	400–1066	1.35V	Up to 16GB	240
DDR4 (DIMM)	1600–3200	800–1600	1.2V	Up to 64GB	288
DDR4 (SO-DIMM)	1600–3200	800–1600	1.2V	Up to 32GB	260
DDR4 (LPCAMM)	1600–3200	800–1600	1.2V	Up to 64GB	LPCAMM
DDR5 (DIMM)	3200–8400	1600–4200	1.05V–1.1V	Up to 128GB	288

DDR5 (SO-DIMM)	3200–6400	1600–3200	1.05V–1.1V	Up to 64GB	260
DDR5 (LPCAMM)	3200–9600	1600–4800	1.05V–1.1V	Up to 128GB	LPCAMM
GDDR5	5000–8000	1250–2000	1.5V	Typically 1–8GB	N/A
DDR6 (projected)	12800+	TBD	TBD	TBD	TBD

LPCAMM

LPCAMM or Low Power CAMM (Compression Attached Memory Module) introduces a different approach to memory capacity, efficiency, and upgradability in modern computing, with its smaller footprint and lower-profile design.

As a result, it’s been widely integrated to power the compact or “thin” laptops

The initial cost is higher, and it is not as widely available as a standard DDR5 SO-DIMM

LPCAMM DDR4

packaging optimizes layout and thermal management, resulting in lower power consumption and improved heat dissipation. This benefits mobile and embedded systems where space and power efficiency are crucial, enabling more compact designs without sacrificing performance. LPCAMM is thus favored in modern laptops and portable devices.

LPCAMM DDR5

LPCAMM packaging for DDR5 supports higher data rates and larger capacities, offering improved signal integrity and lower power consumption through advanced compression and attachment techniques.

This ensures better thermal performance and energy efficiency, essential for high-performance computing. DDR5 LPCAMM modules also provide greater scalability, accommodating future advancements in memory technology and increasing bandwidth and capacity demands in both consumer and enterprise systems.

Overall, LPCAMM advances memory module design by offering superior power efficiency, thermal management, and compact form factors for both DDR4 and DDR5 technologies.

ROM

ROM is a non-volatile storage media used in computer systems and other electronic devices. Its main purpose is to store data that can only be read and not changed, making it easier for different hardware components to communicate with one another. There are different types of ROM, some ROM allow data to be written only once, while others can be rewritten in special ways.

3. Programmable Read-Only Memory (PROM):

- PROM has the ability to program it. Data and instructions cannot be altered after they have been programmed. Hardwiring the ROM chip is not necessary for programming. The data stored in it are permanently stored and can not be changed or erasable. It is used in low-level programs such as firmware or microcode. It means it is the empty PROM chip in which the user can write the desired content or program only once using the special PROM programmer or PROM burner device; after that, the data or instruction cannot be changed or erased.

Advantages of Programmable Read-Only Memory

- Due to the simplicity of its circuitry, this memory is more dependable.
- Today's market is filled with a wide variety of programming software.
- Hardwiring the ROM chip is not necessary for programming.
- Data may be erased and rewritten indefinitely using Programmable read-only memory.

Disadvantages of Programmable Read-Only Memory

- Data of Programmable read-only memory can not be modified.
- Programmable read-only memory often has a finite capacity, and updating it might be costly or challenging.
- Users can not add information frequently and guidelines based on what they need.
- The programmer can burn undesired fuses by selecting a specific path for the current by applying a high voltage.

4. Erasable Programmable Read-Only Memory (EPROM)

EPROM is reorganized. Whereas PROM is more expensive, EPROM is non-volatile. Erasable Programmable Read-Only Memory is for erasing and rewriting the programming. EPROM is more economical than PROM. Reprogrammed of it is limited. Before the era of EEPROM and flash memory, EPROM was used in microcontrollers. It is a non-volatile memory chip that holds data when there is no power supply and can also store data for a minimum of 10 to 20 years. In EPROM, if we want to erase any stored data and re-programmed it, first, we need to pass the ultraviolet light for 40 minutes to erase the data; after that, the data is re-created in EPROM.

Advantages of Erasable Programmable Read-Only Memory

- In Erasable Programmable Read-Only Memory, programme be erased and rewritten.
- The nature of EPROM is non-volatile.
- EPROM is more economical than PROM.
- It Delete all of the prior data in order to reprogram it.

Disadvantages of Erasable Programmable Read-Only Memory

- EPROM cannot erase a single byte on its own. if you delete a single byte, all of EPROM's bytes are also destroyed.
- Data deletion from the EPROM Memory Chip requires extra time.
- Changing the stored data when system needs change is quite costly.
- Users can not add information frequently and guidelines based on what they need.

5. Electrically Erasable Programmable Read-Only Memory (EEPROM)

Electrically Erasable Programmable Read-Only Memory (EEPROM) is a type of non-volatile memory that allows data to be erased and reprogrammed electrically. Data in EEPROM can be erased and rewritten without removing the chip from the device and without the need for special light sources. In EEPROM, the stored data can be erased and reprogrammed up to 10 thousand times, and the data erase one byte at a time.

Advantages of EEPROM

- EEPROM does not need of removed from the device to write new data or erase existing data.
- Data may be erased and rewritten indefinitely using EEPROM technology.
- Electric current is used to upload and erase EEPROM data.
- It is possible to erase data from the EEPROM chip byte.

Disadvantages of EEPROM

- In EEPROM, Different voltages are needed for operations, including recording and erasing data from memory chips.
- Compared to other ROM chips, EEPROM is more costly.
- EEPROM often has a finite capacity, and updating it might be costly or challenging.
- Data stored on EEPROMs has a finite lifespan.

4. Flash Memory

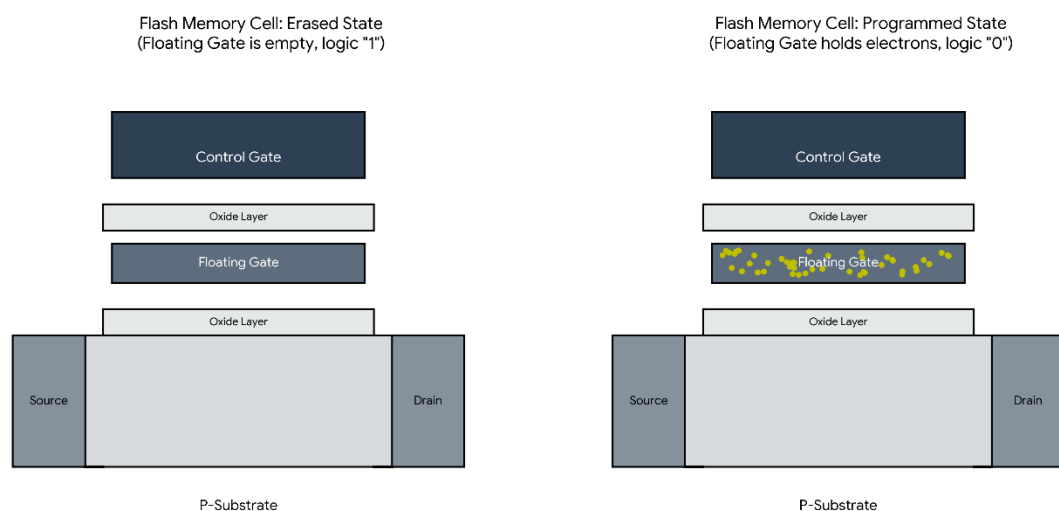
Flash memory is a form of nonvolatile memory with ongoing storage, even without a power source. It enables byte-level rewrites and deletions of data blocks. Unlike traditional hard disk drives (HDDs), which use spinning magnetic disks to store data, Flash memory relies on floating-gate transistors to trap electrons, altering conductivity to stand for data. This technology enables fast access times, low power consumption, and high reliability, making it indispensable in modern computing and electronics.

Working

Flash memory is a type of **EEPROM** chip, which stands for **Electrically Erasable Programmable Read Only Memory**. It has a grid of columns and rows with a cell that has two transistors at each intersection

The two transistors are separated from each other by a thin oxide layer. One of the transistors is known as a **floating gate**, and the other one is the **control gate**. The floating gate's only link to the row, or **wordline**, is through the control gate. As long as this link is in place, the cell has a value of 1. To change the value to a 0 requires a curious process called **Fowler-Nordheim tunneling**.

- **The Control Gate:** Acts as the master switch. Applying voltage here activates the cell to read, write, or erase data.
- **The Floating Gate:** A completely isolated silicon island. Electrons trapped here cannot escape, even when the power turns off.
- **The Erased State (Binary 1):** The floating gate contains **no trapped electrons**. When a read voltage hits the control gate, current flows freely from the Source to the Drain.
- **The Programmed State (Binary 0):** The floating gate is **packed with electrons**. These negative charges create an electric shield that blocks the read voltage, stopping current from flowing to the Drain.



Types of Flash Memory

There are two types of Flash memory most commonly acknowledged: NAND and NOR Flash. NOR Flash was the first of the two to be introduced in 1988 by Intel, while NAND Flash was later introduced by Toshiba in 1989. Their main differences can be identified in their architecture.

[NOR and NAND](#) are named for the way the floating gates of the memory cells that hold data are interconnected in configurations that resemble a NOR or a NAND logic gate.

NOR Flash

NOR Flash is optimized for random access capabilities where it is able to access data in any order and doesn't require following a sequence of storage locations. In terms of its architecture, each of NOR Flash's memory cells are connected in parallel where one end of the memory cell is connected to the source line and the other end is connected to the bit line. This allows the system to access individual memory cells.

NAND Flash

Conversely, NAND Flash is optimized for high-density data storage and gives up the ability for random access capabilities. NAND Flash cells are connected, usually eight memory transistors at a time, in a series to the bit line called a string. Here, the source of one cell is connected to the drain of the next one. This series connection reduces the number of ground wires and bit lines.

In summary, NAND-based Flash memory is ideal for high-capacity data storage, while NOR-based Flash memory is best suited for code storage and execution, generally in small capacities.

Flash Memory Examples

Common examples of Flash memory include:

- Multi-Media Card (MMC) - a Flash-based memory card standard used for solid-state storage in smartphones, digital cameras, music players, video camcorders, and personal computers. These cards store digital information such as text, pictures, audio, and video.
- Solid-State Drive (SSD) - a Flash-based memory used in devices such as computers or video game consoles as a storage device that replaces hard disk drives (HDD) due to its speed and reliability.
- BIOS chip – basic input/output system - is a small memory chip located on a computer's motherboard that stores instructions so the computer can perform basic functions such as booting and keyboard control.
- USB Flash drive - a data storage device that includes Flash memory with an integrated USB interface. This device plugs into a PC, camera, or phone to save or transfer digital data.

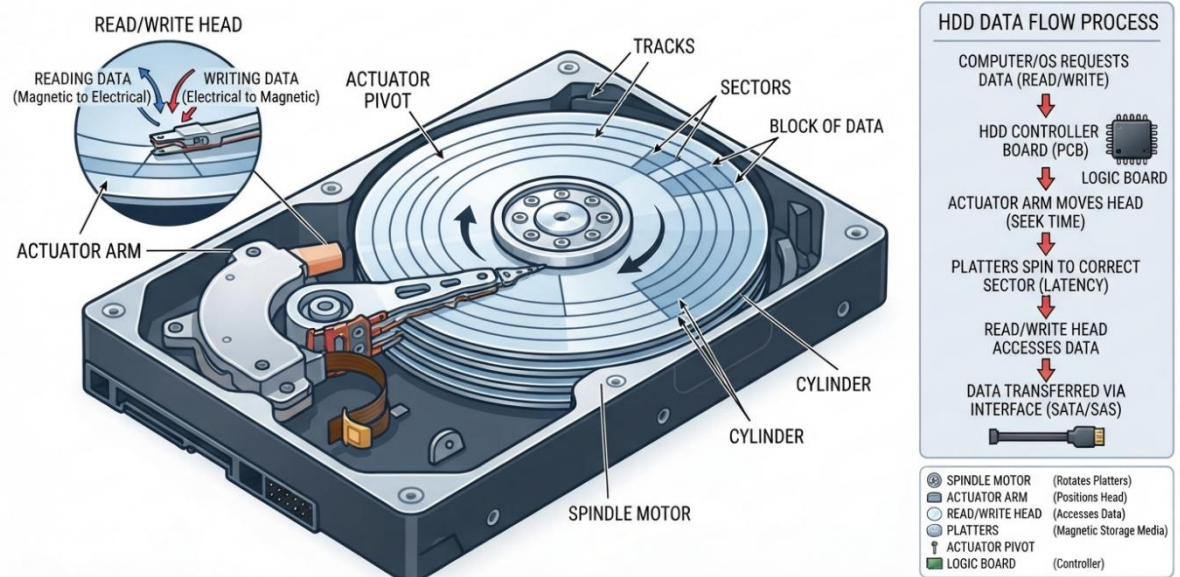
Secondary Storage

Hard Disk Drive

A Hard Disk Drive (HDD) is a mechanical device used to store data permanently. Inside an HDD, it includes:

- **Platters:** Round, shiny metal disks coated with a magnetic layer. This magnetic surface is where actual files are saved. The hard disk platter is what physically contains the data in an HDD and is where an HDD gets its name. The platter in an HDD is rigid, usually made of glass, ceramic or aluminium. These disks are then coated with a thin magnetic layer. This magnetic layer is divided into billions of individual sections, each representing a bit. These sections can then be magnetised to create their own persistent magnetic field; data can be encoded in the states of the magnetic field. Each platter has two sides that can both store data, and each hard disk usually contains between one and five platters.
- **Spindle:** A central pin that spins the platters around at incredibly high speeds (like a spinning CD or record player). The spindle is responsible for holding the platter(s) in place. The spindle is attached to the drive spindle motor, which is responsible for making the platters spin. This is measured in RPM and, for commercial uses, usually varies between 5,400 and 7,200. It is important that the drive spindle motor rotates the disk at a fixed speed and with very little vibration. The spindle motor is connected to feedback circuits that can monitor the speed of the platter; the spindle motor's speed can be controlled to mitigate any changes in the platter speed.
- **Read/Write Heads:** Tiny sensors that float just above the platters. They read data from the disk or write new data onto it. The disk read/write head is used to turn electric current into a magnetic field to manipulate the magnetic field of the platter (writing), or to turn the magnetic field of the platter into an electric current in the read/write head (reading). These read/write heads have an extremely low clearance over the platter, often riding on a thin layer of air, which acts as a lubricant and buffer between the read/write head and the platter. This is known as air bearing. The distance between the platter and the read/write head is known as the flying height. Each platter generally has two read/write heads, one for each side of the platter.
- **Actuator Arm:** A mechanical arm that moves the read/write heads back and forth across the disks. The disk read/write head is attached to a triangular, lightweight alloy called the actuator arm. This arm is moved across the surface of the platter, allowing the read/write head to access different tracks (more on tracks later) by the actuator. By combining the movement of the actuator and the spinning of the platter, the read/write head is able to move across the whole surface of the platter.

DIAGRAM OF HARD DISK DRIVE (HDD) WORKING: COMPONENTS & DATA ORGANIZATION



Working

Step 1: The Platters Spin Permanently

As soon as your computer boots up, an internal motor spins the magnetic platters at a constant, blistering speed (usually 5,400 or 7,200 RPM). Because the disks are always spinning, the drive is always ready to access data instantly without waiting for a motor to ramp up.

Step 2: The Heads Fly on a Cushion of Air

The read/write heads are mounted to the tip of a mechanical actuator arm. The extreme speed of the spinning platters creates a microscopic cushion of air right above the disk surface. The heads literally "fly" on this air cushion, hovering mere nanometers above the magnetic layer without ever physically touching it.

Step 3: Seeking the Right Track

When a file request comes in, the actuator arm uses a powerful electromagnet (called a voice coil motor) to physically sweep across the radius of the disk. It moves back and forth until the read/write head is positioned directly over the specific concentric ring (Track) where the data is stored. This step is called the **Seek Time**.

Step 4: Waiting for the Sector

Once the head is floating over the correct track, the arm locks its position and stands completely still. It waits for the platter's rotation to bring the exact pie-shaped slice (Sector) containing your data directly underneath it. This waiting period is called **Rotational Latency**.

Step 5: Reading or Writing Magnetic Bits

- **Reading Data:** The disk surface contains billions of microscopic magnetic regions mapped out as North or South poles (representing binary 1s and 0s). As these regions pass under the head, they induce a tiny electrical current in the head's sensor. The drive controller converts these electronic pulses back into data your OS understands.
- **Writing Data:** To save a file, the head shoots a precise electrical charge downward. This charge flips the magnetic orientation of the microscopic domains on the sector, permanently recording the new 1s and 0s.

Step 6: Transferring to the System Buffer

The raw magnetic signals are processed by the drive's logic board, temporarily held in a small built-in cache memory buffer, and then sent flying through your interface cables (like SATA) straight to your computer's RAM.

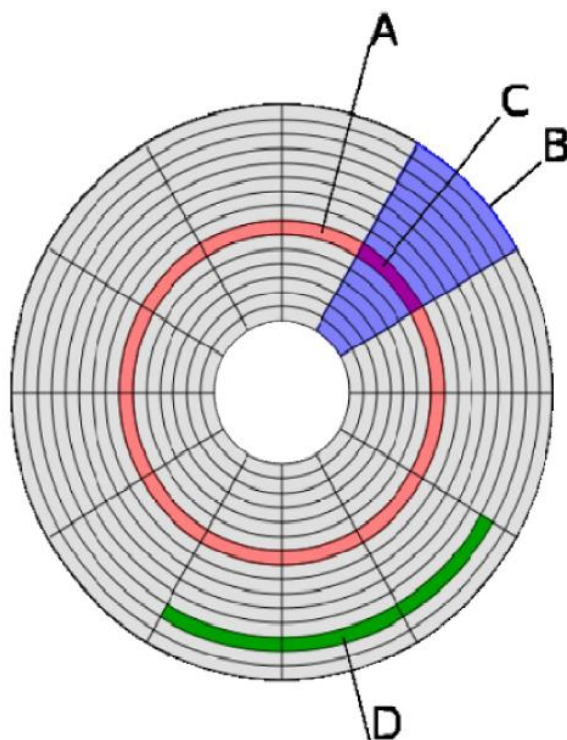
Physical vs Logical Structure

Physical structure	Logical structure
Platters, spindle, heads, actuator, electronics	Tracks, sectors, cylinders, zones and logical block addresses
Describes what exists mechanically/electronically	Describes how storage locations are organized/addressed
Important for construction and operation	Important for storage management and addressing

Logical Structure of HDD

- **Tracks:** Invisible, concentric circles on the platters (like the rings inside a tree trunk). A track is a concentric circular path on a platter surface. Tracks are tightly packed concentric circles used to organize storage and retrieval. Tracks are arranged from the inner region toward the outer edge. A track is divided into sectors. Modern drives can have extremely high track densities, so head positioning is servo-controlled.
- **Sectors:** Small, pie-shaped slices that divide up each track. One sector is the smallest piece of data a computer can read or write at one time. The smallest physical storage block on a disk..Traditionally holds 512 bytes, though modern advanced formats use 4,096 bytes (4K). Grouped to form tracks
- **Cylinders:** A vertical stack of tracks. If you look straight down through all the platters, the tracks that line up perfectly on top of each other form a cylinder. Corresponds to a single position of the read/write head assembly without needing to move the mechanical arm.
- **Zone (Zone Bit Recording / ZBR):** A group of sequential, contiguous concentric tracks on a platter. Outer zones have a larger circumference than inner zones, so they pack more sectors per track to maximise capacity. Keeps data density uniform across the physical surface rather than wasting space on outer tracks.
- A cluster (also called an allocation unit) is the smallest logical amount of disk space that an operating system can allocate to hold a single file. While a sector is the smallest

hardware storage unit on an HDD platter, the operating system groups multiple sectors together into a single cluster to manage file storage more efficiently.



Hard Drive Structure:

A = track

B = sector

C = sector of a track

D = cluster

HDD Interfaces

The supplied PDF describes the drive interface as the physical connection between the drive and the computer's expansion/system bus and lists SCSI, IDE, EIDE, SATA, USB, RAID, IEEE and Solid-State Drive. For modern teaching, these items should be separated into interfaces, historical interfaces, storage architectures and alternative technologies.

Term	Category	Status / modern interpretation
SCSI	Storage interface family	Historical; replaced largely by SAS in enterprise systems.
IDE / PATA	Parallel storage interface	Legacy/obsolete in modern PCs.
EIDE	Enhanced/extended PATA terminology	Legacy/obsolete.

SATA	Serial storage interface	Still widely used for internal HDDs; SATA Revision 3.x provides up to 6 Gb/s link rate.
SAS	Serial enterprise storage interface	Widely used for enterprise HDDs; supports dual-port and enterprise features.
USB	External host interface	Common for external HDD enclosures and portable drives.
RAID	Storage architecture	Not an interface; combines multiple drives for performance, capacity and/or redundancy.
IEEE/FireWire	Historical external interface family	Used historically for storage; largely replaced by USB/Thunderbolt and other modern interfaces.
SSD	Alternative storage technology	Not an HDD interface; uses flash memory.

Capacity and Technology

The figures below are current manufacturer-listed examples

Manufacturer/product family	Current listed example	Technology/position
Seagate Exos X / Mozaic	36 TB Mozaic; 32 TB and 30 TB Exos X also listed	Enterprise/high-capacity; current product page lists Mozaic 36 TB and Exos X 32/30/25 TB.
Western Digital WD Gold	Up to 26 TB	Enterprise SATA; 7,200 RPM; CMR; 512 MB cache on listed high-capacity models.
Western Digital Ultrastar DC HC690	Up to 32 TB	Enterprise/data-centre; 11-disk platform; SMR/ePMR/UltraSMR variants.
Seagate BarraCuda 3.5	Up to 24 TB	Desktop SATA family; 24 TB model listed.
WD general internal HDD portfolio	Up to 28 TB on selected models	High-capacity internal HDD portfolio; selected-model figure.

SSD

SSD (Solid-State Drive) is a **non-volatile secondary storage device** used to store operating systems, applications, documents, videos, and other digital data.

Unlike an HDD, an SSD has **no rotating platters, spindle motor, or moving read/write heads**. It stores data electronically using **NAND Flash memory**.

Working Principle of SSD

An **SSD (Solid-State Drive)** stores data electronically in **NAND Flash memory**. Unlike an HDD, it has **no rotating platters, spindle motor, or moving read/write heads**.

The working of an SSD involves the **host interface, SSD controller, Flash Translation Layer (FTL), NAND Flash memory, error correction, wear leveling, and garbage collection**.

The Storage Medium: NAND Flash Memory

The foundation of an SSD is **NAND flash memory**, a type of non-volatile storage that retains data even when the computer is powered down.

Floating Gate / Charge Trap Transistors: The smallest unit of an SSD is the memory cell. These cells use a process called **quantum tunneling** to push electrons through an insulating barrier into a "floating gate" or "charge trap".

Representing 1s and 0s: When electrons are trapped inside the gate, they block electrical current, representing a binary **0**. When the gate is empty, current flows freely, representing a binary **1**.

Memory Structure: Cells are organized like a grid. They are grouped together into **Pages** (typically 4KB to 16KB), which are further grouped into **Blocks** (usually 128 to 512 pages).

Read, Write, and Erase Data

- **Read Operations:** The drive checks the voltage level of a cell to see if electrons are trapped. This is incredibly fast and can be done at the individual **page level**.
- **Write Operations:** The drive applies a higher voltage to inject electrons into specific cells. Data can only be written to empty, individual **pages**.
- **Erase Operations:** Cells cannot simply be overwritten. To change data, the existing data must be wiped out first. However, due to the electrical design, an SSD can only erase data in whole **blocks**, not individual pages.

SSD Controller

SSDs function through an arrangement of NAND flash memories. The more NAND flash memories in a unit, the more storage an SSD has and the faster its read and write speeds are. A "channel" connects multiple flash memories to the controller. The controller handles data, transfer speeds, and data integrity and retention. Within the memory controller is a series of

processes called wear leveling, garbage collection, and bad block management, all of which control the health and longevity of the SSD.

Advantages of SSDs

- **Speed**
SSDs offer exceptional speed due to their lack of moving parts. This results in faster boot times, quicker file transfers, and smoother overall system performance. For instance, the CL6 M.2 SSD, known for its high-speed performance, provides an outstanding example of how SSD technology has evolved, making data access almost instantaneous.
- **Durability**
SSDs are more resistant to physical shock and vibration compared to HDDs. The absence of moving components makes them less susceptible to damage caused by drops or accidental impacts, ensuring data integrity and device longevity. Products like CL6 M.2 SSD are designed with durability in mind, offering robust protection for your stored data.
- **Energy Efficiency**
SSDs require less power to operate, making them more energy efficient. This advantage contributes to extended battery life and improved overall efficiency, particularly for laptops and portable devices. For those seeking a balance between performance and energy consumption, CL6 M.2 SSD can be an ideal choice.

SATA SSD

- Uses the SATA interface (same as older hard drives)
- Maximum speed: **~550 MB/s**
- Common form factors:
 - **2.5-inch SATA SSD** (most common)
 - **mSATA** (older laptops)

Best for: Upgrading older desktops and laptops.



2. NVMe SSD (PCIe)



- Uses the PCIe interface with the NVMe protocol
- Much faster than SATA SSDs
- Typical speeds:
 - PCIe Gen 3: **2,000–3,500 MB/s**
 - PCIe Gen 4: **5,000–7,500 MB/s**
 - PCIe Gen 5: **10,000–14,000+ MB/s**

Common form factors:

- **M.2 2280** (most common)
- **PCIe add-in card (AIC)** for desktops

Best for: Gaming, video editing, programming, and high-performance computing.

3. External SSD



- Connects via USB or Thunderbolt
- Portable and plug-and-play
- Typical speeds:
 - USB 3.2: **500–2,000 MB/s**
 - Thunderbolt: **Up to 2,800+ MB/s**

Best for: Backups, portable storage, and transferring large files.

Differences Between HDD and SSD

No.	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Storage Technology	Stores data magnetically on rotating platters.	Stores data electronically using NAND flash memory.
Moving Parts	Has moving parts such as platters, spindle and read/write heads.	Has no moving parts .
Speed	Relatively slower because data access involves mechanical movement.	Much faster because data is accessed electronically.

No.	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Access Time	Higher access time due to seek and rotational latency.	Very low access time because there is no mechanical movement.
Noise	Produces noise due to rotating platters and moving heads.	Operates silently.
Power Consumption	Generally consumes more power because the platters must rotate.	Generally consumes less power, especially compared with mechanical HDDs.
Shock Resistance	More sensitive to physical shock and vibration.	More resistant to shock and vibration.
Cost	Cheaper per GB and economical for large-capacity storage.	Usually more expensive per GB, although prices continue to decrease.
Data Recovery	Some mechanical failures may be recoverable by specialist services.	Recovery can be difficult after controller or NAND failure, depending on the architecture.
Best Use	Best for bulk storage, backups, archives and large amounts of data.	Best for operating systems, applications, gaming and high-performance computing.

Difference Between DVD and USB

DVD (Digital Versatile Disc) and **USB (Universal Serial Bus)** are both used for storing and transferring digital data, but they work in very different ways. A DVD uses **optical storage**, while a USB flash drive uses **flash memory**.

No.	DVD (Digital Versatile Disc)	USB Flash Drive
Storage Technology	Uses optical technology to store data on a disc.	Uses NAND flash memory to store data electronically.
Data Reading/Writing	Data is read using a laser beam in an optical drive.	Data is read and written electronically through a flash-memory controller.

No.	DVD (Digital Versatile Disc)	USB Flash Drive
Capacity	A single-layer DVD normally stores about 4.7 GB ; dual-layer DVD can store about 8.5 GB .	Available in much larger capacities, commonly from 8 GB to 1 TB or more .
Speed	Relatively slow because the disc must rotate and data is accessed optically.	Generally much faster because data is accessed electronically.
Rewritability	DVD-R is write-once, while DVD-RW/DVD+RW can be rewritten.	Can normally be erased and rewritten many times.
Portability	Thin and lightweight, but requires a protective case to prevent scratches.	Very compact, lightweight and easy to carry.
Physical Parts	Requires an optical disc and a DVD drive for reading/writing.	Contains flash memory and a controller with no moving parts.
Connection/Interface	Uses an optical disc drive connected internally or externally to the computer.	Connects directly through a USB port .
Durability	Can be damaged by scratches, heat and physical bending.	Generally more resistant to scratches and normal physical handling.
Reusability	Depends on the type; read-only and write-once DVDs cannot be reused for new data.	Designed for repeated data writing and deletion.
Compatibility	Requires a computer or device with a compatible DVD drive.	Requires a compatible USB port; widely supported by modern computers and other devices.
Power Requirement	The DVD drive requires power to rotate and operate the laser mechanism.	USB supplies both power and data communication through the port.
Data Transfer	Primarily designed for distributing or storing fixed amounts of data.	Designed for convenient, repeated transfer of files between devices.

No.	DVD (Digital Versatile Disc)	USB Flash Drive
Security	A finalized/write-protected DVD can be useful for maintaining an unaltered copy.	Data can easily be modified or deleted unless protected by software or hardware.
Modern Usage	Mainly used for movies, software distribution, backups and archival purposes where optical media is appropriate.	Commonly used for file transfer, portable storage, installation media and bootable drives.

Cache memory levels (L1, L2 and L3)

CPU cache, also known as cache memory, is a small high-speed memory located inside or near the processor. CPU cache is a small, high-speed memory integrated into the processor. It stores frequently accessed data and instructions so the [CPU](#) can retrieve them without waiting for main memory. Acting as a middle layer between the processor and [RAM](#), the cache helps maintain smooth performance, faster load times, and efficient multitasking even under heavy workloads.

The Role of CPU Cache

The main purpose of CPU cache is to reduce the time it takes for the processor to access data. Since RAM is much slower than the CPU, direct memory access would constantly interrupt processing. Cache memory keeps a ready copy of the data the CPU is likely to need next, allowing it to continue executing instructions without delay.

In everyday use, this means faster response when launching applications, switching between browser tabs, or loading complex files. By keeping frequently used data close to the cores, cache memory allows the processor to work continuously rather than waiting for information to travel from system memory.

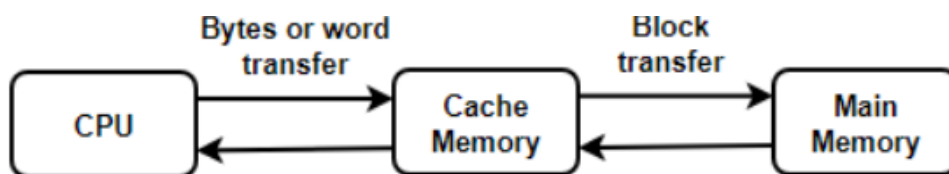


Figure 4.6: Cache Memory

How Caching Works in Processors

When a processor needs data, it first checks the cache.

- If the required data is **found in the cache**, it's called a **cache hit**. The CPU can use the data immediately, ensuring smooth and uninterrupted processing.

- If the data **isn't found**, it's a **cache miss**. The processor must then fetch the data from RAM, which takes more time. Once retrieved, a copy is stored in the cache for future use.

This hit-and-miss system allows the cache to “learn” and predict what the CPU will need next. Over time, it minimizes repeated trips to RAM and keeps execution pipelines filled with data. In modern CPUs, this mechanism works across multiple levels—L1, L2, and L3—each balancing speed, size, and proximity to the core. Together, they ensure that critical data is always close at hand, maintaining high performance across gaming, creative, and professional workloads.

Level	Location	Typical Size	Speed	Function
L1 Cache	Inside each CPU core	32–128 KB	Fastest	Immediate data access
L2 Cache	Near the core	256 KB–1 MB	Very fast	Stores recently used data
L3 Cache	Shared across cores	4–64 MB	Fast	Coordinates data sharing

L1 Cache – Fastest and Closest to the Core

The **L1 cache** is the first and fastest layer of memory within the CPU. It sits directly inside each processor core and handles the most frequently accessed data and instructions. Because it operates at the same speed as the core, it delivers results within nanoseconds.

L1 cache is usually split into two parts:

- **Instruction cache (L1i)** – stores CPU instructions.
- **Data cache (L1d)** – stores values the CPU needs for calculations.

Its small capacity—typically between 32KB and 128KB per core—means it can only hold the most critical information, but its speed ensures minimal delay in task execution.

L2 Cache – Balancing Speed and Capacity

The **L2 cache** is the second level and acts as a bridge between L1 and L3. It's larger, usually between 256KB and 1MB per core, but slightly slower. L2 stores data that isn't currently in L1 but might be needed again soon.

This layer reduces the number of cache misses that require data to be pulled from RAM. When L1 doesn't contain the requested data, the CPU automatically checks L2. Its moderate size and proximity to the core make it a critical part of balancing fast access and efficient storage.

L3 Cache – Shared for Multi-Core Coordination

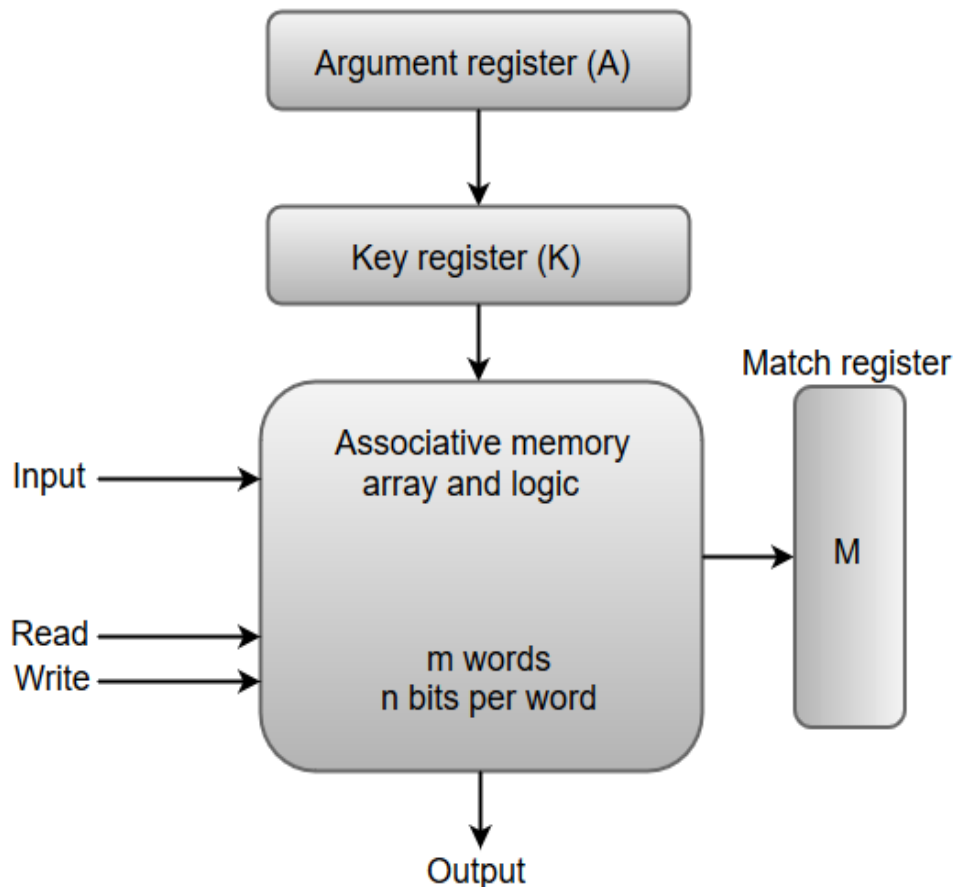
The **L3 cache** sits farther from the core and is shared among all cores within the processor. It's much larger—often between 4MB and 64MB—but slower than L1 and L2. Its primary function is to coordinate data sharing and reduce redundancy across cores.

When multiple cores process different threads of the same application, L3 ensures they can access shared data without repeatedly retrieving it from RAM. This shared structure improves efficiency in workloads like video editing, 3D rendering, and gaming, where multiple cores must work on related data simultaneously.

Associative Memory

- A Memory unit is accessed by its content or data is called content-addressable memory (CAM) or associative storage or associative array.
- It is a special type of memory that is optimized for performing searches through data, as opposed to providing a simple direct access to the data based on the address.
- It can store the set of patterns as memories when the associative memory is being presented with a key pattern, it responds by producing one of the stored pattern which closely resembles or relates to the key pattern.
- When a word is written in an associative memory. No address is given.
- The memory is capable of finding an empty unused location to store a word.

The following diagram shows the block representation of an Associative memory.

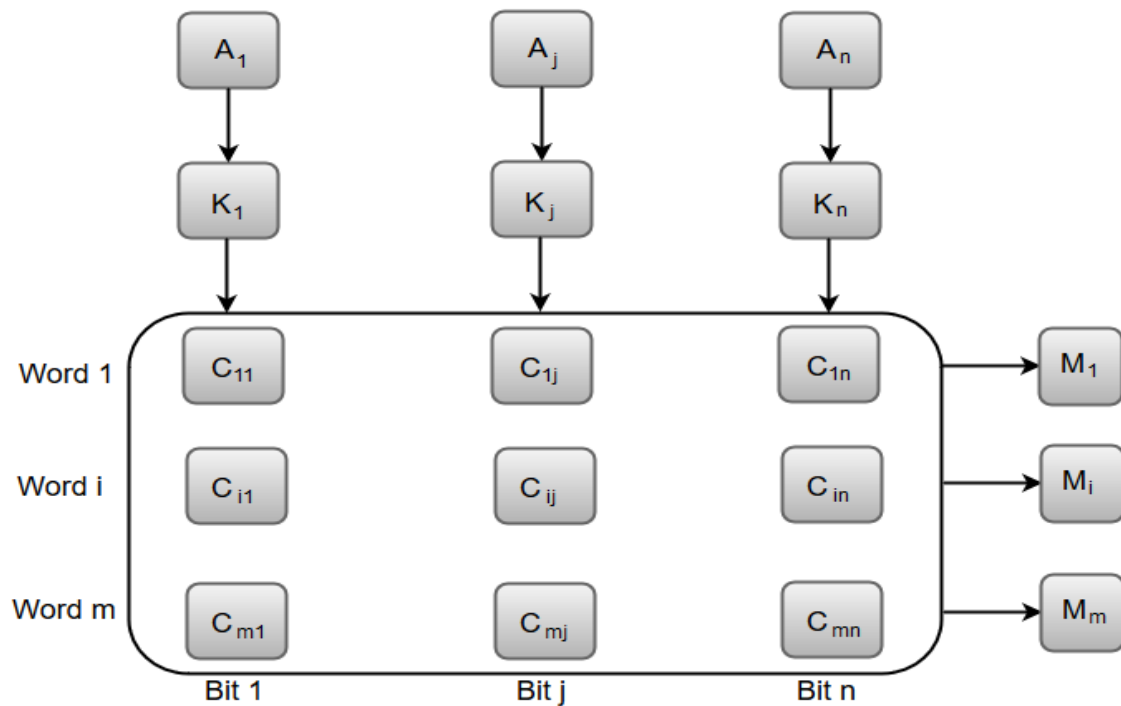


A	101 111100	
K	111 000000	
Word 1	100 111100	no match
Word 2	101 000001	match

- **Argument Register:** It contains words to be searched. It contains 'n' number of bits.
- **Match Register:** It has m-bits, One bit corresponding to each word in the memory array. After the making process, the bits corresponding to matching words in match register are set to '1'.
- **Key Register:** It provides a mask of choosing a particular field/key in argument register. It specifies which part of the argument word need to be compared with words in memory.
- **Associative Memory Array:** It combines word in that are to be compared with the arguments word in parallel. It contains 'm' words with 'n' bit per word.

The following diagram can represent the relation between the memory array and the external registers in an associative memory.

Associative memory of m word, n cells per word:



Applications of Associative memory:-

- It can be only used in memory allocation format.
- It is widely used in the database management systems, etc.
- Networking: Associative memory is used in network routing tables to quickly find the path to a destination network based on its address.
- Image processing: Associative memory is used in image processing applications to search for specific features or patterns within an image.
- Artificial intelligence: Associative memory is used in artificial intelligence applications such as expert systems and pattern recognition.
- Database management: Associative memory can be used in database management systems to quickly retrieve data based on its content.

Advantages of Associative memory :-

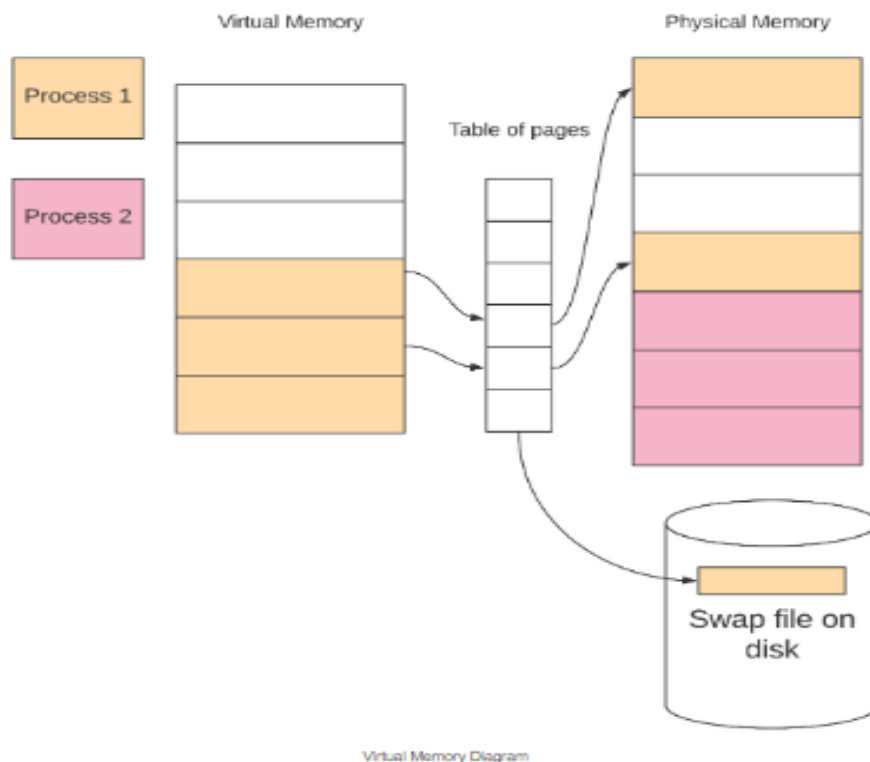
- It is used where search time needs to be less or short.
- It is suitable for parallel searches.
- It is often used to speedup databases.
- It is used in page tables used by the virtual memory and used in neural networks.

Disadvantages of Associative memory :-

- It is more expensive than RAM
- Each cell must have storage capability and logical circuits for matching its content with external argument.

Virtual Memory

- Virtual memory is the part of a hard drive and is extra memory that works with a combination of RAM (Random Access Memory). RAM is also known as the main memory or physical memory.
- Virtual memory is the partition of logical memory from physical memory.
- Virtual memory is a memory management technique used by operating systems to give the appearance of a large, continuous block of memory to applications, even if the physical memory (RAM) is limited.
- A virtual memory is what its name indicates- it is an illusion of a memory that is larger than the real memory.
- This partition supports large virtual memory for programmers when only limited physical memory is available.
- Virtual memory can give programmers the deception that they have a very high memory although the computer has a small **main memory**.
- The speed of the main memory is very fast as compared to virtual memory.



- Suppose you are running some applications of windows e.g. you are running MS Word, Chrome, and Calculator.
- If for some time you minimize MS Word then this application is inactive and the data of MS Word will be transferred from main memory to virtual memory. The movement of data from main memory to virtual memory is known as a swap out of the process.
- Main memory has a limit of space and it can only process a limited number of processes at a time. But virtual memory gives extra space for the processes which are larger to be occupied in the main memory.
- When space in main memory exceeds the limit then unused processes are moved from RAM to virtual memory. When the CPU needs those inactive processes then they are moved from virtual memory to RAM this movement of processes is known as a swap-in of process.

The virtual memory paging process uses page tables, which translate the virtual addresses that the OS and applications use into the physical addresses that the MMU uses. Entries in the page table indicate whether the page is in RAM. If the OS or a program does not find what it needs in RAM, then the MMU responds to the missing memory reference with a page fault exception to get the OS to move the page back to memory when it is needed. Once the page is in RAM, its virtual address appears in the page table.

Segmentation is also used to manage virtual memory. This approach divides virtual memory into segments of different lengths. Segments not in use in memory can be moved to virtual memory space on the drive. Segmented information or processes are tracked in a segment table, which shows if a segment is present in memory, whether it has been modified and its physical address. In addition, [file systems](#) in segmentation are only made up of segments that are mapped into a potential address space of a process.

Segmentation and paging differ as an amount of memory model in terms of how memory is divided; however, the processes can also be combined. When combined, memory gets divided into frames or pages. The segments take up multiple pages, and the virtual address includes both the segment number and the page number.

Other page replacement methods include first in, first out (FIFO), optimal algorithm and least recently used (LRU) page replacement:

- The FIFO method has memory select the replacement for a page that has been in the virtual address for the longest time.
- The optimal algorithm method selects page replacements based on which page is unlikely to be replaced after the longest amount of time; although difficult to implement, this leads to fewer page faults.
- The LRU page replacement method replaces the page that has not been used for the longest time in the main memory.

Advantages of virtual memory

- By using virtual memory many applications or programs can be executed at a time.
- Main memory has limited space but you can increase and decrease the size of virtual memory by yourself.
- Users can run large programs that have a size greater than the main memory
- The data which is common in memory can be shared between RAM and virtual memory
- CPU utilization can be increased because more processes can reside in the main memory
- The cost of buying extra RAM is saved by using virtual memory

Disadvantages of virtual memory

- Virtual memory tends to slow down your computer because the speed of the hard drive is less than RAM
- Switching between applications takes more time

- As virtual memory allocates space on the hard drive, the user's hard drive space becomes less
- The overall performance of the system is affected
- The algorithm for virtual memory is difficult to implement